ITC INFOTECH TECHNICAL ROUND INTERVIEW QUESTIONS 2025

1. HR / Personal Section

Covers identity checks, personality, motivation, and communication questions.

- Show / take ID proof, face screenshot, surroundings check
- Self introduction
- About family, native place, schooling, hobbies, strengths, weaknesses
- Projects and internships overview
- Why we should hire you / Why this company / Why this role
- Preferred language and rating
- Career goals (after 5 or 10 years)
- Dream company and favorite person
- What are your challenges / how do you handle pressure
- Situational questions (team issues, deadline handling, conflict resolution)
- Any questions from your side
- Tell about ITC Infotech / About your college
- Role in team project
- Rate yourself in languages
- Why Python and not Java
- If given opportunity, where you see yourself
- Do you know cooking?
- How are you / how do you celebrate Diwali

---

2. Project & Internship Section

Questions about your real or academic work.

- Explain your project in detail
- Technologies used and why chosen
- Outcomes, learnings, and challenges
- Role in the project
- Web deployment and API usage
- Dataset sources, preprocessing

- Internship role, domain, and duration
- Approved by college or self-project
- When deadline is near, how do you manage?
- LLM and Small Language Models
- Cloud computing and its applications
- Prompt engineering
- Unit testing vs Integration testing
- Debugging process in project context

---

3. Python Programming Section

Covers core Python language features and theory.

- __init__() usage
- Mutable vs Immutable objects
- Difference between == and is
- Docstring and its usage
- Python built-in datatypes
- Iterators and Iterables
- Decorators (definition, syntax, example)
- Generators
- Conditional statements and loops
- Functions and recursion
- File operations, context managers, with statements
- Exception handling (try, except, finally)
- Modules vs Packages
- Memory management in Python
- GIL (Global Interpreter Lock)
- Self keyword, indentation
- PEP8
- Deep vs Shallow copy
- Debugging in Python
- Numpy, Pandas, Missing values in datasets

- Python libraries
- Python interpreter
- Sequence data types
- Difference between list, tuple, set, dictionary
- Tuple vs List (speed, mutability)
- Slicing and examples
- String buffer and string builder
- Python multithreading and multiprocessing
- Singleton
- Monkeys patching, yield keyword
- Does Python support multiple inheritance?
- Zip function
- Metadata in Python

---

4. Python Coding Section

Direct coding questions or logic explanation.

- Palindrome program
- Even or Odd program
- Fibonacci series (first 10 numbers)
- Factorial program
- Reverse string (with or without built-in functions)
- Remove duplicates from a list
- Anagram check for two strings
- String concatenation (based on your name)
- Prime number (true/false)
- Maximum of 3 numbers
- Infinite loop syntax
- Merging two dictionaries
- Swapping variables
- Check vowels in name (True/False)
- Pattern printing program

- Biggest of 2 numbers without if-else
- Reverse of number (logic only)

---

5. Java Programming Section

Covers Java OOPs, syntax, and key concepts.

- What is JVM, JRE, JDK
- Platform independence in Java
- Constructor (default, parameterized)
- Method overloading and overriding
- Access modifiers (private, protected, public, default)
- Class and Object
- Four pillars of OOPs — Inheritance, Encapsulation, Polymorphism, Abstraction
- Abstract class and Interface
- Static keyword and programs
- Difference between Array and ArrayList
- LinkedList and ArrayList
- Exception and finally keyword
- Garbage collection in Java
- Threads and multithreading
- Deadlock
- Lambda expression
- String pool and class loader
- Call by value vs call by reference
- Copy constructor
- Java program: palindrome, even/odd, second largest number
- Difference between pre-increment and post-increment
- Difference between do-while and while
- Java platform features and syntax questions

---

6. C and C++ Programming Section

- Write name in C without semicolon

- Data types and declaration
- Pointers and structures
- Arrays vs structures
- Memory leaks and management
- Difference between C and C++
- Operators in C
- Call by value/reference
- Shallow vs Deep copy
- Basic structure of C program
- Infinite loop using do-while
- Copy constructor
- C++ vs Java
- OOPs in C++

---

7. Database / SQL Section

Covers all SQL and database theory.

- What is DBMS / RDBMS
- Primary key, Foreign key, Unique key
- DDL, DML, DCL commands
- Create table, database
- SQL Constraints
- Joins (inner, left, right)
- DISTINCT, LIKE, UNION, UNION ALL
- Delete vs Drop vs Truncate
- Indexes in SQL and MongoDB
- Sorting and select top 5 rows
- SQL query for 98th employee
- 2-table join query
- Group By
- Normalization types
- SQL Data types

- ACID properties
- Difference between SQL and MongoDB
- Documents in MongoDB
- Dataframes and operations in Python
- File automation and storage
- Schema meaning

---

8. Web Development Section (HTML, CSS, JS, React, API)

Front-end and web-related questions.

- HTML structure, tags, and semantic tags
- Block vs Inline elements
- Difference between id and class
- CSS types and usage
- Flexbox
- JavaScript basics, var, let, const
- Difference between frontend and backend
- Modules in JavaScript
- REST API and HTTP methods
- SPA (Single Page Application)
- Event delegation, synchronous vs asynchronous
- React.js basics
- Webpack, AIDL, CORS, Tags
- MEAN / MERN stack
- Troubleshooting code on server vs system

---

9. Computer Networks Section

- HTTP vs HTTPS
- IP and IP classes
- UDP
- DNS, DHCP, SMTP
- TCP layers

- SNMP, SSH protocols
- Client-server model
- Difference between HTTP and HTTPS
- Debugging network-related errors

---

10. Operating Systems & Concurrency

- Threads and Multithreading
- Multiprocessing vs Multithreading
- Deadlock
- Stack vs Heap memory
- Process synchronization
- OS role in program execution

---

11. Object-Oriented Programming (Conceptual)

Common to Java, Python, and C++.

- Explain OOPs and its four pillars
- Class, Object
- Inheritance types
- Polymorphism (compile-time, run-time)
- Encapsulation and Abstraction
- Access modifiers
- Constructor and Destructor
- Interface and Abstract Class
- Overloading and Overriding
- Instance variables vs local variables
- SDLC (Software Development Life Cycle)

---

12. Data Structures & Algorithms Section

- Stack vs Queue
- Array vs ArrayList
- LinkedList

- Linear vs Non-linear data structures
- Binary tree
- Sorting algorithms (bubble, linear sort)
- Pattern printing
- Binary search
- Big-O concept (implicit from problem-solving)

---

13. Aptitude & Logical Reasoning Section

- Age and probability problems (father-son, colored balls)
- Train speed and time questions
- Water jug (3L, 5L to get 4L) puzzle
- Logical reasoning (days of week, statements on shapes)
- Angle between hour and minute hand
- Biggest among two numbers
- Common aptitude-based riddles

---

14. Emerging Tech / Miscellaneous Section

- AI, ML, DL, Cloud computing
- Computer Vision, OpenCV, Mediapipe, CNN
- Cybersecurity
- Debugging meaning and process
- Prompt Engineering
- Testing: Unit testing, Integration testing
- Bug meaning and handling
- SDLC
- Activity in app development
- Keras, TensorFlow
- AI pros and cons
- Impact of AI on future jobs